Feasibility Validation Report: Agricultural Compliance Platform

Executive Summary

Feasibility Assessment: FEASIBLE

Confidence Level: Medium-to-High (conditional on Weeks 1-4 partnership formalization and schema lock)

Key Finding: Native iOS/Android with SQLite offline-first architecture is technically sound and differentiating. Single-user MVP scope, extension service data partnership, and immutable logs eliminate major complexity risks. Feasibility hinges on three critical path gates: (1) formalize extension service partnerships by Week 4, (2) lock EPA database schema by Week 2, (3) stress-test offline sync with real applicators starting Week 9.

MVP Timeline: 14-16 weeks to pilot-ready (Weeks 5-8 build, Weeks 9-12 pilot with 20-25 applicators, Weeks 13-16 analysis and seed pitch prep).

Go/No-Go Gates: If extension service partnerships aren't formalized by Week 4 or EPA schema isn't locked by Week 2, timeline extends 3-4 weeks. If either slips, reassess before proceeding.

Technical Feasibility

Core Architecture: - Stack: Native iOS (Swift) + Android (Kotlin), SQLite local database with sync-on-connectivity - Complexity: Medium (offline sync is proven but requires rigorous testing) - Proven: Yes (Figma, Salesforce, and others use this pattern successfully)

Key Technical Decisions: - Immutable logs post-submission (applicators cannot edit after logging; eliminates sync conflicts and compliance liability) - Single-user MVP (defer contractor multi-user features to Phase 2) - Simple schema: User → Applications → EPA Rules. No complex relationships or multi-field data models - Extension service partnership for initial EPA RUP data; quarterly scraper updates in Phase 2; AI-driven rule-scanning in Phase 2

Technical Risks & Mitigations: - Offline sync edge cases (network drops mid-sync, storage limits, version conflicts): Allocate 2-3 weeks QA testing starting Week 9 with real applicators, not lab-only - EPA rule complexity exceeds estimates: Validate simplest compliance case (Can applicator X apply chemical Y in state Z on date D?) with extension partners by end Week 2 - Database schema rework: Lock schema by end Week 2 after partner input; build migration logic to avoid mid-development refactoring

Scalability: MVP schema supports 1000+ applicators without redesign; offline sync tested to 500+ applications per device. Post-MVP, revisit for contractor multi-user scenarios.

Operational Feasibility

Key Operations: - Pilot support (Weeks 9-12): 15 hrs/week founder + email/Slack support for 20-25 applicators. ~40 min per applicator over 4 weeks (realistic) - Onboarding: Extension services provide warm intros + 30-min training; you handle follow-up questions - Metrics tracking: Daily active users, offline vs. wifi usage, willingness to pay, re-activation intent

Resource Requirements: - Team: 1 senior iOS dev (Weeks 1-16), 1 senior Android dev (Weeks 1-16), 1 backend/data engineer (Weeks 1-8), founder for partnerships + pilot support (15 hrs/week Weeks 9-12) - Tools: Firebase (auth, backend), GitHub (version control), Slack (team comms), Figma (design) - Extension service data access and quarterly updates

Bottlenecks & Solutions: - Extension service partnership delays: Formalize in writing Weeks 1-4; weekly check-ins to track recruitment pipeline. Target 50% recruited by Week 7 - Offline sync testing: Start with real applicators Week 9, not Week 14. Allocate 4 weeks for edge-case discovery and fixes - Support scaling: Build pilot playbook (FAQ, troubleshooting, onboarding guide) in Weeks 1-4 to reduce per-applicator support time

Build vs. Buy Analysis

Build: - Native iOS/Android apps: Core differentiation. Offline-first architecture not available in off-the-shelf compliance tools - SQLite + sync logic: Proven pattern; free. Implement with built-in iOS/Android sync libraries - EPA RUP quarterly scraper: Low complexity; validates data quality before Phase 2 AI integration - PDF generation: Build lightweight (iText for Android, PDFKit for iOS). Simple compliance report export

Buy/Partner: - Weather API: Dark Sky (Apple) or WeatherAPI. Cost $0.50-2/user/month (negligible). Reliable source - Maps SDK: MapKit (iOS) or Google Maps (Android). Built-in SDKs; no licensing complexity - Authentication: Firebase Auth or AWS Cognito. Production-ready; handles compliance and security - EPA RUP data (MVP): PARTNER with extension services for validated baseline; build scraper later - AI rule-scanning (Phase 2): Evaluate AWS Textract or Claude API to scan EPA PDFs and flag rule changes

Integration Complexity: Medium. Firebase + native SDKs are well-documented. EPA data integration straightforward (quarterly CSV/JSON import). No major third-party dependency risks.

Compliance & Regulatory

Regulatory Requirements: - EPA RUP compliance guidance: App is advisory, not legal counsel. Immutable logs provide historical compliance records. Applicators responsible for verifying with EPA.gov - Data validation: Extension service partners provide written confirmation that initial EPA RUP dataset is accurate (mitigates liability for bad guidance) - Rule change responsiveness: AI-driven scanning (Phase 2) detects EPA updates; you can push rule changes between quarterly releases if critical changes occur mid-season

Data Privacy: - GDPR: Minimal impact (mostly EU). If you expand internationally, review data transfer agreements - CCPA: Applies if you have California users. Recommend opt-in cloud sync (logs stored locally by default; sync to cloud is optional). Include CCPA privacy notice in app - Farm operation data: Keep sensitive data (chemical types, field locations, applicator identity) local-first. Encryption at rest (iOS Keychain, Android Keystore) required

Industry-Specific: - Compliance liability: Clear disclaimers in app ('This app is advisory; consult EPA.gov for authoritative guidance'). Rule change log showing 'Last Updated: [date]' transparency. Immutable logs protect you from retroactive compliance claims - Legal review recommended: Budget $2-5K for ag lawyer to review liability disclaimers, data privacy practices, and contractor data handling (before Phase 2). Do this before seed pitch to show diligence to investors

Expert Debate Insights

Areas of Consensus: - Native iOS/Android with SQLite is sound architecture for offline-first use case - Immutable logs (no post-submission editing) eliminate sync complexity and clarify compliance liability - Single-user MVP scope is correct; contractor features (Phase 2) are risk reduction - Extension service partnerships are critical path; must be formalized in writing by Week 4 - 14-16 week timeline is realistic if schema locked by Week 2 and sync testing starts Week 9

Areas of Debate & Resolutions: - EPA rule complexity: Initial debate on whether rules could exceed engineering estimates. Resolution: Start with simplest case (single state, single chemical, single license type) and validate with extension partners by Week 2. Expand rule logic incrementally - Contractor feature in MVP: Debate between Phase 1 vs. Phase 2. Resolution: Defer to Phase 2 because problem interviews showed solo applicators are viable beachhead; contractors are Phase 2 expansion - Support burden realism: Question on whether 15 hrs/week is sufficient for 25 applicators. Resolution: Confirmed as realistic if extension services do warm intros and initial training; you handle follow-up only

Unresolved Questions: - Will extension services actively promote app post-pilot, or just provide initial introductions? (Mitigated by written agreements in Week 1-4) - What is acceptable EPA rule accuracy threshold before launch? (Recommend partner validation + disclaimer approach; revisit with legal counsel) - How quickly can you pivot if offline sync testing reveals fundamental architectural flaw? (Have contingency: Phase 2 could add web-based alternative if mobile sync proves untenable)

Implementation Roadmap

Phase 1 - MVP (Weeks 1-8, Build-to-Pilot): - Weeks 1-4 (Partnerships & Schema): Formalize extension service agreements in writing. Design and lock EPA RUP database schema with partner input. Build pilot playbook (onboarding, FAQ, support runbook) - Weeks 5-8 (Development): Implement native iOS/Android apps with SQLite offline sync. Integrate EPA RUP data. Add weather API, GPS/maps, PDF export. Firebase auth setup. Internal testing only - Deliverables: Pilot-ready iOS + Android apps, schema documentation, partnership agreements, pilot playbook

Phase 2 - Pilot (Weeks 9-12, Real-World Validation): - Weeks 9-12: Onboard 20-25 applicators via extension service intros. Founder provides 15 hrs/week email/Slack support. Real-world offline sync stress testing. Track metrics (DAU, offline usage, feature adoption, willingness to pay) - Deliverables: Sync testing logs, pilot metrics, applicator feedback, conversion data

Phase 3 - Analysis & Seed Prep (Weeks 13-16, Decision Gate): - Weeks 13-16: Analyze pilot data (15%+ conversion target, 50%+ re-activation intent target). Recalculate CAC, LTV with real numbers. Prepare seed pitch deck. Make go/no-go decision on full product build - Deliverables: Pilot analysis report, seed pitch deck, go/no-go recommendation

Critical Path Items: - Extension service partnership formalization (Week 1-4): Delays cascade to all downstream phases - EPA schema design + partner validation (Week 1-2): Rework mid-development costs 3-4 weeks - Offline sync testing with real users (Week 9+): Lab testing insufficient; real-world stress test essential - Pilot recruitment follow-up (Week 5-8): Weekly checkins with extension services to hit 50% recruited by Week 7

Recommendations

Immediate Actions (This Week): 1. Schedule formalization meetings with 2 extension services: Secure written commitments on data access, 10-15 applicator introductions by Week 9, revenue share/co-marketing model, pilot timeline 2. Sketch EPA RUP database schema: Identify simplest compliance case to validate in Week 2 (Can X apply Y in Z on date D?). Document state-by-state rule variations 3. Identify dev team: Lock 1 senior iOS dev + 1 senior Android dev before Week 1. Timeline risk is high if you can't staff immediately

Before MVP Build (Weeks 1-4): - Formalize extension service partnerships in writing (non-negotiable go/no-go gate) - Lock EPA schema after partner input; document migrations strategy - Build pilot playbook: onboarding guide, FAQ, support runbook, troubleshooting guide - Set up Firebase, GitHub, Figma, development infrastructure - Sketch compliance checking logic; identify complexity gaps early

During MVP Build (Weeks 5-8): - Weekly sync testing on development devices (don't wait for Week 9)

- Partner review of EPA data ingestion pipeline (Week 4-5) - Internal beta testing with team members using offline + online scenarios - Prepare marketing/onboarding materials for pilot launch

During Pilot (Weeks 9-12): - Aggressive metrics tracking: daily active users, offline vs. wifi split, willingness to pay (explicit ask Week 11), likelihood to renew - Weekly sync testing logs documenting every failure and fix - Weekly check-ins with extension services on recruitment status - Collect qualitative feedback: What features matter? What's friction? Would you pay?

Team Priorities: 1. Partnership formalization (founder/leadership) — blocks everything 2. EPA schema design (senior engineer + partners) — blocks development 3. iOS/Android implementation in parallel (dev team) — critical path 4. Offline sync testing (QA + real users) — risk mitigation 5. Pilot playbook & support infrastructure (operations) — scaling readiness

Go/No-Go Decision Gates

Gate 1: Week 4 (Partnership Formalization) - Requirement: 2+ extension services commit in writing to data access, 10-15 applicator introductions by Week 9, defined revenue share <20% - If met: Proceed to MVP build (Week 5) - If not met: Pivot channel strategy (direct sales, different extension services, or pause MVP build until partnerships are locked)

Gate 2: Week 2 (Schema & Complexity Validation) - Requirement: EPA RUP database schema locked. Simplest compliance case (Can X apply Y in Z?) validated with partners. No hidden complexity discovered - If met: Proceed with confidence to full development - If not met: Extend schema design phase 1-2 weeks; push pilot start to Week 10

Gate 3: Week 9 (Pilot Recruitment) - Requirement: 50% of target applicators (10-12 of 20-25) recruited and ready to onboard - If met: Begin pilot as planned - If not met: Delay pilot 2 weeks; adjust recruitment targets downward to 15-20 applicators

Gate 4: Week 16 (Conversion & Seed Readiness) - Requirement: 15%+ pilot-to-paid conversion (3-4 applicators paying by Week 16) OR 50%+ expressed re-activation intent + clear path to revenue - If met: Proceed to seed pitch with traction data - If not met: Extend pilot 4 weeks or pivot business model before fundraising